Drone Traffic Fingerprinting

What We Can Pick Up, Exploit, and Package

Source research: Christof (2021) DJI Wi-Fi Protocol RE thesis · Behind the Wings (2023) DJI Enhanced Wi-Fi hijacking · OcuSync DroneID analysis (2026) · Detecting Drones via Encrypted Traffic Analysis (2019) · Drone Profiling through Wireless Fingerprinting (2018)

The Problem with the Current Paper

The IJRPR paper's 4-factor model (OUI + packet rate + RSSI variance + SSID) works for basic detection but leaves massive gaps:

	Risk	Impact
r vendor ID	MAC randomization	Drone with random MAC = 40 points lost
c pattern analysis	Encrypted traffic indistinguishable from any high-bandwidth client	High false-positive rate in urban environments
only	5 GHz drones invisible	Misses newer DJI Air/Mavic on 5.8 GHz
ode	No direction finding	Cannot distinguish drone 50m away from drone 20
col-level analysis	Cannot differentiate drone type or flight mode	No threat discrimination

Layer 1 — What the ESP32 Picks Up Today (No Changes Needed)

These signals are available from promiscuous 802.11 frame headers alone. No decryption needed. No firmware change required for the basic capture — just smarter processing.

Signal	Visible In	What It Tells You
Source MAC	Every frame	Device identity (can be randomized)
RSSI	rx_ctrl struct	Range estimate, variance tracking
Frame type	Frame Control field	Management vs data vs control
Packet size	Raw frame length	Biometric: small commands vs large video
Direction	SA vs DA	Who is sending — drone or controller
Channel	Set by firmware	Which 2.4 GHz channel is active
Beacon SSID	Beacon frame body	Drone AP name (if not hidden)

Layer 2 — Traffic Statistics (Encrypted but Classifiable)

These signals survive WPA2 encryption. Multiple academic papers demonstrate 90%+ detection accuracy from these features alone, using as few as 170 packets (820ms of traffic).

Feature	Drone Signature	Detectability
Packet size distribution	Bi-modal: 56B controller → 1472B drone	Trivial — size histogram per MAC
Traffic direction ratio	Drone transmits 10-50× more than controller	Byte counter per direction
Video frame cadence	~33ms intervals (29.4 fps H.264)	Timing histogram of drone→controller pkts
Burst structure	4-6 MTU frames back-to-back then pause	Consecutive MTU-size frame counter
Flow continuity	No idle gaps >200ms in active sessions	Max inter-packet gap per device
Channel utilization	40-80% duty cycle on single channel	Time between frames on channel

Key insight: A smartphone streaming video to YouTube also sends large packets, but it sends both ways (ACKs, control). A drone link is almost entirely one-directional from drone to controller. This asymmetry alone is a powerful classifier.

Layer 3 — Decrypted DUML Protocol (Post-WPA2 Key)

Critical finding from Christof (2021): DJI's Wi-Fi protocol has **no application-layer encryption.** The DUML payload is plaintext once WPA2 is decrypted. This means:

Protocol Element	Offset	What We Can Extract
Content type byte	Wi-Fi Hdr +0x06	Classifies every packet as command vs video vs camera
0x55 delimiter	DUML Hdr +0x00	Firm identifier of DJI drone traffic at payload level
Command-set + command	DUML Hdr +0x09/+0x0A	Identifies exact operation (fly/camera/battery/etc.)
Telemetry data	DUML Body, Cmd 0x43/Set 0x03	Latitude, longitude, height, velocity, pitch/roll/yaw, flight status
Battery data	DUML Body, Cmd 0x03/Set 0x0D	Cell voltage, estimated remaining power
Session ID	Wi-Fi Hdr +0x02	Tracks all packets in one flight session
Packet counter	Wi-Fi Hdr +0x24	Sequential ordering — detect packet loss or injection
CRC-16 Kermit	DUML Trailer	Non-standard CRC — verifiable protocol signature
Video stream	UDP payload, content type 0x02	H.264, 1280×720, ~30fps — extractable via FFmpeg

What We Can Exploit — Asymmetric Advantages

These are the angles where the defender has inherent advantage over the drone operator:

1. The drone cannot stop transmitting.

A drone in flight requires continuous bidirectional communication. It cannot go silent without initiating return-to-home or landing. This means the detection target is always emitting — no windowing, no intermittent operation.

2. The video stream is a forcing function.

To stream 1080p video at 30fps, the drone must send ~4 Mbps of H.264 data. This requires MTU-sized frames (~1472 bytes) at a sustained rate. No consumer IoT device has this traffic profile. A phone streaming video comes close, but phones have bidirectional ACK traffic — drones don't.

3. The protocol structure is deterministic.

The DJI DUML protocol has fixed delimiters (0x55), known CRC algorithms (CRC-16 Kermit), and fixed field positions. Even without decryption, a 0x55 search at expected offsets within encrypted packets provides a statistical signature for DJI drone traffic.

4. The hardware is cheap.

An ESP32 costs $5. It can already capture all the data needed for Layers 1 and 2 analysis. Adding Layer 3 analysis requires only capturing the WPA2 handshake (same $5 hardware, software addition). This is 1,000× cheaper than commercial C-UAS sensors.

5. The attacker cannot spoof the traffic profile.

To evade traffic-pattern detection, a drone would need to mimic a different device class — changing packet sizes, timing, and directionality. This would require modifying the drone's firmware. Consumer drones don't allow this. Only custom-built FPV drones could attempt it.

What We Can Package — Detection Layers

Proposed five-layer detection model, building on the paper's 4-factor approach:

Layer	What It Detects	Hardware	Conf. Weight
1 — MAC OUI + SSID	Known drone vendor prefixes + drone AP names	ESP32	20 pts
2 — Traffic Direction Ratio	Drone sends 10-50× more than it receives	ESP32	25 pts
3 — Packet Size Distribution	Bi-modal 56B / 1472B signature	ESP32	25 pts
4 — Video Frame Cadence	~33ms timing intervals in drone→controller flow	ESP32	20 pts
5 — DUML Protocol Signature (post-WPA2)	0x55 delimiter + CRC-16 Kermit at known offsets	ESP32 + key capture	10 pts

Proposed weight redistribution vs the original paper:

Original: OUI(40) + Rate(30) + Var(20) + SSID(10) = 100

Proposed: DirRatio(25) + PktSize(25) + OUI(20) + Timing(20) + Var(5) + SSID(5) = 100

This reduces reliance on MAC OUI (which can be randomized) and elevates traffic-pattern features (which cannot be spoofed without firmware modification).

FPV & Homebrew Drones — Where the ESP32 Goes Blind

The analysis above applies primarily to **consumer Wi-Fi drones** (DJI in Wi-Fi mode, Parrot, Autel, MAVLink-based drones with Wi-Fi telemetry). FPV and homebrew drones use completely different RF

— and the ESP32 cannot see them.

FPV Drones — Invisible to ESP32

ystem	Protocol	Frequency	Detectable By
ol link	ExpressLRS, Crossfire/CRSF, ELRS	900 MHz or 2.4 GHz FHSS	HackRF, BladeRF — not ESP32
link	Analog NTSC/PAL	5.8 GHz analog	RTL-SDR + downconverter, or RF power de
link	Digital (DJI FPV, Walksnail Avatar, HDZero)	5.8 GHz OFDM	SDR + protocol-specific decoder
etry (optional)	MAVLink over serial (not Wi-Fi)	915/868 MHz (SiK radio)	RTL-SDR (narrowband FM)
nk	PWM/PPM/SBUS over serial	Sub-GHz proprietary	Not easily detectable

The critical point: An FPV drone emits NO 802.11 frames. The ESP32 sniffer is completely blind to it. The video link on 5.8 GHz is the only reliable passive detection vector — and it requires different hardware.

However: FPV drones still have a video link that must transmit continuously while flying. That 5.8 GHz signal has its own distinctive traits — OFDM preamble signatures for digital links, NTSC sync pulses for analog. These are detectable with a HackRF or even a simple RF power detector tuned to 5.8 GHz.

Homebrew Drones — Paradoxically MORE Detectable When on Wi-Fi

A homebrew drone with a Raspberry Pi or ESP32 companion computer running MAVLink over Wi-Fi is actually **easier to identify** than a DJI drone:

Trait	MAVLink Drone (Wi-Fi)	DJI Drone (Wi-Fi Mode)
Protocol	Open-source, documented	Proprietary, reverse-engineered
Transport	MAVLink over UDP	DUML over UDP
Heartbeat	1Hz, always-on, fixed msg ID #0	No heartbeat packet — continuous telemetry stream
Message IDs	Known (GPS=33, ATTITUDE=30, BATTERY=147, etc.)	Known (telemetry=0x43/0x03, battery=0x03/0x0D)
Packet sizes	Variable, per message type	Fixed clusters per content type
Encryption	None (MAVLink is plaintext by design)	WPA2 + no app-layer encryption
Detection ease	Trivial — any Wi-Fi sniffer can decode	Requires WPA2 key capture or statistical analysis

Key insight: MAVLink's 1Hz heartbeat is the most reliable drone signature in existence. Every MAVLink drone broadcasts its system ID, component ID, and flight mode every second — in plaintext, over UDP. Any Wi-Fi sniffer within range can identify a MAVLink drone with 100% certainty from a single packet.

Drones Without Remote ID — The Real Detection Challenge

Remote ID (FAA Part 89 / ASTM F3411) broadcasts identification over Wi-Fi or BLE. Drones that don't broadcast Remote ID fall into these categories:

	Remote ID?	Detectable By ESP32?	Primary Detection Vector
(Wi-Fi)	Yes (FAA-compliant)	Yes — Remote ID broadcast + Wi-Fi AP	ESP32 + 5.8 GHz Rx for OcuSync path
(OcuSync)	Yes (embedded in OcuSync)	Not via Wi-Fi — DroneID in SDR signal	HackRF (DroneID decodable per luyii-c
hantom 4, Spark)	No (pre-RID firmware)	Yes — Wi-Fi AP is active	ESP32 (traffic pattern analysis)
ew (ELRS + analog)	No (unless module added)	No — no Wi-Fi at all	5.8 GHz RF power detector + SDR
Pi (MAVLink)	Optional (can add RID module)	Yes — MAVLink UDP stream	ESP32 (MAVLink heartbeat is 100% ID
with FCC module	Depends on firmware	Maybe — depends on telemetry radio	Multi-modal

The hardest detection problem: An FPV drone on ELRS (900 MHz FHSS) with analog video (5.8 GHz NTSC). No Wi-Fi, no BLE, no Remote ID. Detection requires a 5.8 GHz RF energy detector (cheap, but gives only presence, not classification) or an SDR with FHSS tracking capability (expensive, high compute). This is the gap that a $5 ESP32 cannot fill.

Multi-Modal Detection Strategy — Covering All Drone Types

	Detects	Cost	Coverage Gap
promiscuous)	Consumer Wi-Fi drones, MAVLink drones, Remote ID broadcasts	$5	FPV drones, OcuSync-only drones.
SDR (2.4 GHz)	OcuSync DroneID, FHSS control links	$30-300	5.8 GHz video links, requires DSP d
wer detector	Analog/digital FPV video (always-on)	$10-50	No classification — presence only
8 GHz downconverter	Digital FPV video signatures (OFDM)	$50-100	Requires protocol-specific decoder
combined node	Wi-Fi drones + FHSS + 5.8 GHz energy	$50-100	Full spectrum coverage in single no

Recommended minimum viable stack for broad coverage: One ESP32 (for Wi-Fi drones, MAVLink, Remote ID) + one $10 RTL-SDR dongle with a 5.8 GHz antenna (for FPV video presence detection) + one HackRF (for OcuSync DroneID decode). Total BOM: ~$50-150 per node.

Implementation Path

e	What We Build	Tim
e 1: Traffic Stats	Add per-MAC packet size histogram + direction ratio to ESP32 firmware. Weight 50% of confidence score.	Wee
e 2: Timing Analysis	Add inter-packet timing histogram. Detect 30fps cadence. Weight 20% of score.	Wee
e 3: WPA2 Key Capture	Log 4-way handshake from promiscuous frames. Store PMKID for offline decrypt.	Wee
e 4: DUML Parser	Post-process captured PCAPs. Extract telemetry + video from decrypted flows.	Wee
e 5: Real-time ML	TinyML classifier on ESP32 using traffic features. 5-layer model in firmware.	Mor

Key References

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3. Sciancalepore et al. (2019). *Detecting Drones Status via Encrypted Traffic Analysis.* ACM WiSec. Demonstrates drone classification from encrypted packet features.

4. Bisio et al. (2018). *Unauthorized Amateur UAV Detection Based on WiFi Statistical Fingerprint Analysis.* IEEE. 90%+ detection accuracy from traffic statistics.

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